

STEP 4 — Google Drive (Output)

Deliverable ຂອງ STEP 4 ຕາມ Book 3 (Prompt) + Book 2 Phase 4. ສ້າງອີງ Book 1: §3.2, §9, §13, §45, §46, §54. ຂຽນ src/DriveService.gs (Repository ຝັງ Drive) — ຄອບຄຸມ Folder, Upload, Move, Archive, RecycleBin, Thumbnail, Cleanup.
△ STEP 3 Approved (99.8/100). DriveService ຮຽກ DriveApp ໄດ້ບ່ອນດຽວ (§22.1) — Service/UI ຫ້າມຮຽກໂດຍກົງ.

1. Folder & File Naming Convention (§10, §3.2)

ໂຄງສ້າງ Folder ຮູບຂອງລູກຄ້າ ແລະຮູບແບບຊື່ໄຟລິມາດຕະຖານ:

Photos/{Year}/{Month}/{Category}/{CustomerCode}/

ຕົວຢ່າງ: Photos/2026/07/event/CUS-000001/

File name: {CustomerCode}_{id}.{ext}

ຕົວຢ່າງ: CUS-000001_PHOTO-000012.jpg

Base Folder	ໜ້າທີ່
Photos	ຮູບຖ່າຍສະບັບ (ຕາມ Year/Month/Category/Customer)
Thumbnails	cache ຂອງ Preview
Documents	ເອກະສານ (Phase 10)
Temp	upload ຊົ່ວຄາວ — cleanup 24 ຊມ.
RecycleBin	soft-delete — ລຶບຖາວອນ 30 ວັນ
Archive	ຂໍ້ມູນເກົ່າ
Backups	backup Sheets/ໄຟລີ (§41)
System	Templates, Logo, Config

△ ensureBaseFolders() ສ້າງ base folders ຄົບຖ້ວນດ້ວຍຕອນ setup. Root = ROOT_FOLDER_ID (PropertiesService §13).

2. Drive Error Codes & Retry Policy

ຂະຫຍາຍ Error Code ຈາກ §54 ໃຫ້ຄອບຄຸມການເຮັດວຽກກັບ Drive:

Code	ຄວາມໝາຍ	Retry?
E002	Upload / Drive op failed	ໃຊ້ (ສົ່ງສູດ 3 ຄັ້ງ, backoff)
E003	Drive Quota Exceeded	ບໍ່ (ຢຸດ + ແຈ້ງ Admin)
E004	Folder / Root Missing	ບໍ່ (ສ້າງ folder ຫຼືກວດ config)
E005	Permission Denied	ບໍ່ (ກວດ Sharing/Role §13)
E006	Invalid File Type	ບໍ່ (ກວດ ALLOWED_EXTENSIONS)
E007	File Too Large	ບໍ່ (ກວດ MAX_UPLOAD_MB)

Retry Policy (`_withRetry`): ລອງໃໝ່ສູງສຸດ 3 ຄັ້ງ ພ້ອມ backoff 400ms × attempt. E003 (quota) ບໍ່ retry. ຖ້າຄົບ 3 ຄັ້ງຍັງລົ້ມ → throw E002.

3. DriveService.gs API

Function	ໜ້າທີ່
<code>ensureBaseFolders()</code>	ສ້າງ base folders ຄົບຕາມ §3.2
<code>getCustomerPhotoFolder(code, cat, when)</code>	ຫາ/ສ້າງ Photos/{Y}/{M}/{Cat}/{Customer}/
<code>buildFileName(code, id, ext)</code>	ຊື່ໄຟລ໌ມາດຕະຖານ {code}_{id}.{ext}
<code>uploadPhoto(blob, opts)</code>	validate (E006/E007) + upload + thumbnail + retry
<code>generateThumbnail(fileId)</code>	ຄົ້ນ URL thumbnail (Drive built-in)
<code>moveToArchive(fileId)</code>	ຍ້າຍໄປ Archive (§45)
<code>moveToRecycleBin(fileId)</code>	soft-delete → RecycleBin (ຄູ່ Database.softDelete)
<code>cleanupTemp()</code>	ລຶບ Temp ເກົ່າ > 24 ຊມ.
<code>cleanupRecycleBin()</code>	ລຶບຖາວອນ RecycleBin > 30 ວັນ
<code>createCleanupTriggers()</code>	ຕັ້ງ time-driven trigger ໃຫ້ cleanup ອັດຕະໂນມັດ

4. Cleanup Job (§45, §46)

Cleanup ແລ່ນເປັນ time-driven trigger (Background) ບໍ່ໃຫ້ User ລໍຖ້າ:

- `cleanupTemp` — ທຸກມື້ເວລາ 02:00: ລຶບໄຟລ໌ Temp ທີ່ `getLastUpdated` ເກົ່າກວ່າ 24 ຊົ່ວໂມງ.
- `cleanupRecycleBin` — ທຸກມື້ເວລາ 03:00: `setTrashed` ໄຟລ໌ RecycleBin ທີ່ເກົ່າກວ່າ 30 ວັນ.
- ຕັ້ງຄັງດຽວ: `run createCleanupTriggers()` (ໃຊ້ `ScriptApp.newTrigger`).

5. Deliverables & Acceptance (STEP 4)

Deliverables

- DriveService.gs (Folder/Upload/Move/Archive/Recycle/Thumbnail/Cleanup)
- Folder & File Naming Convention
- Drive Error Codes + Retry Policy
- Cleanup Job (time-driven triggers)

Acceptance (§25 + Phase 4)

- ສ້າງ Folder ອັດຕະໂນມັດຖືກຕ້ອງຕາມ Customer Code ✓

- Upload validate (E006/E007) + retry ✓
- ຄືນ Error Code ມາດຕະຖານເມື່ອລົ້ມເຫລວ (§54) ✓
- ບໍ່ Share Folder ສາທາລະນະ (§13) ✓
- Upload ≤ 5s (§56) — ວັດໃນ Phase 9/14

6. ສະຫຼຸບ (Code Review) & ວິທີທົດສອບ

ໄຟລ໌ທີ່ເພີ່ມ: src/DriveService.gs (Repository ຝັ່ງ Drive) — syntax ຜ່ານ node --check.

ວິທີທົດສອບ (ໃນ **GAS**): (1) setConfig ROOT_FOLDER_ID. (2) ensureBaseFolders() → ກວດ folders. (3) uploadPhoto(blob, {customerCode,category,id}) → ກວດໄຟລ໌+thumbnail. (4) moveToRecycleBin/moveToArchive. (5) Negative: ໄຟລ໌ .exe → E006; ໄຟລ໌ໃຫຍ່ → E007. (6) createCleanupTriggers() ແລ້ວກວດ trigger.

ຂັ້ນຕໍ່ໄປ: ຫຼັງ Accept STEP 4 → STEP 5 (Authentication) — Auth.gs (Login/Role).